

Dear Editor-in-Chief and Associate Editor,

Please find enclosed our manuscript entitled “**Between recent speciation and macroevolutionary patterns: gene discordance within phylogenomics and integrative taxon-omics of Apollo butterflies (Papilionidae: Parnassiinae)**” for consideration for publication as a ‘Research Article’ in *Systematic Biology*.

In this manuscript, we investigate whether the imbalance in lineage diversification within a clade, showing both rapid and recent speciation for one part of the tree versus slower and older speciation in another part of the tree, would result in different gene discordance dynamics (between/within mitochondrial and nuclear genes) and show a link between micro- and macro-evolutionary scales. To address this topic, we present the most comprehensive phylogenomic analysis of the relationships and boundaries of Apollo butterflies (Papilionidae: Parnassiinae), which shows such a tree imbalance with a rapid and recent speciation (genus *Parnassius*, ~60 species) and a slower and older radiation (seven other genera, ~20 species). We sequenced 145 *de novo* genomes using whole-genome sequencing with both short- and long-reads techniques and integrated these with publicly available genomes to represent 98% of all species currently described. At the macroevolutionary scale, we performed thorough phylogenomic inferences and compared gene discordance between clades. At the microevolutionary scale, we investigated the impact of this gene discordance using species delimitation and population genomics analyses.

We found that genus *Parnassius* went through an episode of rapid early radiation, which is currently ongoing, and such a radiation correlates with low gene concordance and difficulties in making taxonomic decisions. On the contrary, the other and less diversified genera show a significantly higher level of gene concordance, which results in easier decisions to draw boundaries between species. The example of Parnassiinae therefore suggests a link between recent microevolutionary processes and macroevolutionary patterns, where rapid radiation can cause both uncertainties in deep-time phylogenomic inferences and problems in species delimitation in recent times.

While gene concordance is being extensively studied, the exploration of gene discordance during low and rapid diversification at macro- and microevolutionary scales with a nearly complete species sampling remains rare in the field. With the increasing availability of genomic data, cases similar to *Parnassius* with high gene discordance and difficulties to delineate species boundaries will be increasingly unveiled and studied, which can add a great amount of information on the underlying biological processes governing ancient and recent speciation events. This study provides a guideline for other groups using a combination of analyses based on genomes from macroevolutionary (phylogenomics) to microevolutionary (species delimitation analyses, population genomics) scales.

Given the broad scope of our analyses and the general question we address in this study, we think that our findings will be of strong interest to readers working in evolutionary biology, phylogenetics, genomics, and integrative taxonomy. We thank you for your consideration and hope that the novelty of our results will make our study a worthy contribution to *Systematic Biology*.

Yours sincerely,

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